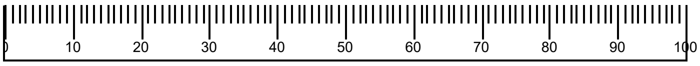
Wesnoth Stitch — stitch-symbol legibility test (#20)

This sheet exists to answer one question the code cannot: **do all 37 chart symbols stay distinguishable when printed at the size a real chart is read at?** §5.3 flags three marginal pairs — `C/G`, `E/F`, `P/R` — and notes that the rules behind the set were validated on a screen, never on paper. If a pair fails here, that glyph is removed and the colour cap `MAX_COLOUR_COUNT` drops by one: the two numbers are the same number.

1 · Check the scale before you read anything

Print at **100% / Actual Size**, not "fit to page". Then measure the bar below. It must be exactly 100 mm. If it is not, the print was scaled and every judgement on the following pages is void.



Measured length: ____________ mm

2 · What scale is being tested, and why

Not the fabric's. A chart is read at whatever size its grid lands on the page. A 72-cell Wesnoth sprite printed across A4's ~170 mm of printable width gives cells of about **2.36 mm**, which puts each glyph at roughly **4.8 pt** — that is where §5.3's "legible at ~5 pt" comes from, and it is the row to judge. The sheet brackets it: 1.81 mm is the stitched size on 14-count Aida (deliberately unfair), 3.00 mm is a chart given more room.

3 · How to record your answers

Page 2 is a reference — just look. Page 3 is the drill: for each pair, tick *distinct* only if you can tell them apart *without* comparing them side by side. Page 4 is a blind test: write what you see, and do not turn to the key on page 6 until you are done. Page 5 is two real charts.

Rendered through Chromium with the app's font stack (`DejaVu Sans, Segoe UI Symbol, Arial, sans-serif`). The PDF export (§5.5) does not exist yet, so this tests glyph *shapes* at print size, not the eventual embedded font. Resolved font is reported at the foot of page 6.

Page 2 · The whole set, at four scales

All 37 glyphs, in chart order (most distinctive first — a low-colour chart only ever spends the top of this list). Glyph height is 0.72 × the cell, exactly as the app draws it.

1.81 mm cells 14-count Aida — the stitched size itself. Unfairly small.

● ■ ▲ ◆ ○ □ △ ◇ ☆ ◐ + × # = A B C D E F G H I J K L M N P R S T U V W Y Z

2.12 mm cells a 80-cell chart across A4

● ■ ▲ ◆ ○ □ △ ◇ ☆ ◐ + × # = A B C D E F G H I J K L M N P R S T U V W Y Z

2.36 mm cells a 72-cell sprite across A4 — the real case

● ■ ▲ ◆ ○ □ △ ◇ ☆ ◐ + × # = A B C D E F G H I J K L M N P R S T U V W Y Z

3.00 mm cells a chart given more room

● ■ ▲ ◆ ○ □ △ ◇ ☆ ◐ + × # = A B C D E F G H I J K L M N P R S T U V W Y Z

Page 3 · The confusion drill

Three marginal pairs and five controls the design already resolved. The fifth column is the honest test: the same two glyphs with three others between them, which is how a chart presents them.

	1.81 mm	2.36 mm	3.00 mm	separated, 2.36 mm	at 2.36 mm, verdict
C / G marginal (§5.3)	C G	C G	C G	C ▲ M □ G	☐ same ☐ distinct
E / F marginal (§5.3)	E F	E F	E F	E ▲ M □ F	☐ same ☐ distinct
P / R marginal (§5.3)	P R	P R	P R	P ▲ M □ R	☐ same ☐ distinct
● / ◆ control — solid blob rule	● ◆	● ◆	● ◆	● ▲ M □ ◆	☐ same ☐ distinct
○ / ◇ control — outline	○ ◇	○ ◇	○ ◇	○ ▲ M □ ◇	☐ same ☐ distinct
+ / × control — strokes	+ ×	+ ×	+ ×	+ ▲ M □ ×	☐ same ☐ distinct
# / = control — strokes	# =	# =	# =	# ▲ M □ =	☐ same ☐ distinct
◐ / ● control — half fill	◐ ●	◐ ●	◐ ●	◐ ▲ M □ ●	☐ same ☐ distinct

If any of **C/G**, **E/F**, **P/R** reads *same*, name it here — each one removed lowers the colour cap from 37 by one:

____

Page 4 · Blind identification

Write the glyph you see (draw it, or name it: "solid circle", "capital G"). Do not look at the key on page 6 first. Errors here matter more than the drill above — this is the task a stitcher actually performs.

1.81 mm cells — numbers 1–12

1	2	3	4	5	6	7	8	9	10	11	12
H	R	☆	G	+	W	☆	☆	◆	C	I	O

1. __________ 2. __________ 3. __________ 4. __________ 5. __________ 6. __________ 7. __________ 8. __________ 9. __________
10. __________ 11. __________ 12. __________

2.36 mm cells — numbers 13–24

13	14	15	16	17	18	19	20	21	22	23	24
B	U	A	C	☆	G	S	▲	K	C	K	D

13. __________ 14. __________ 15. __________ 16. __________ 17. __________ 18. __________ 19. __________ 20. __________
21. __________ 22. __________ 23. __________ 24. __________

3.00 mm cells — numbers 25–36

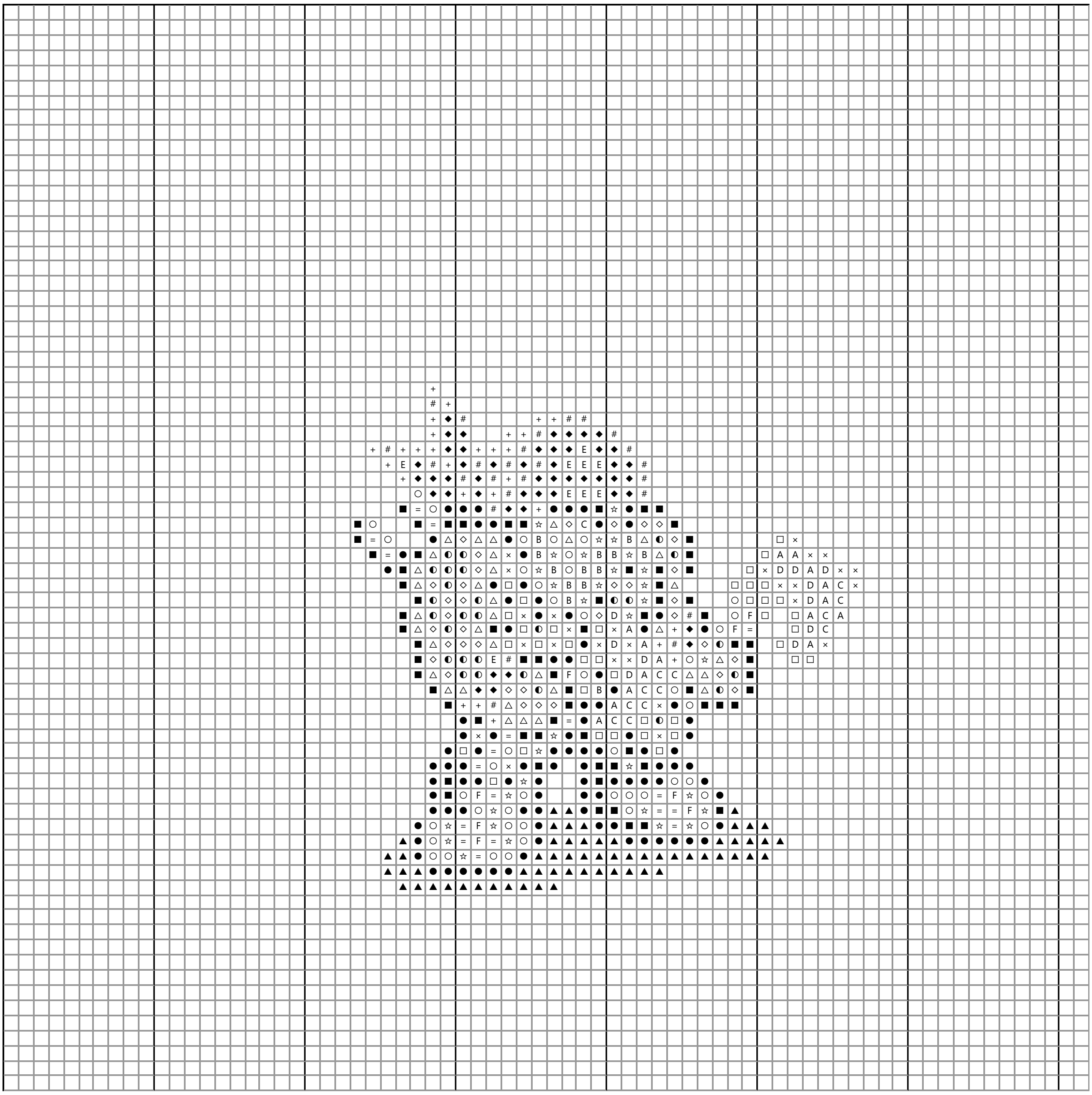
25	26	27	28	29	30	31	32	33	34	35	36
B	P	U	E	V	E	I	◐	#	☆	C	M

25. __________ 26. __________ 27. __________ 28. __________ 29. __________ 30. __________ 31. __________ 32. __________
33. __________ 34. __________ 35. __________ 36. __________

Page 5 · Two real charts, at the true printed scale

Dwarvish scout, k = 20 — a typical sprite, comfortably under the cap

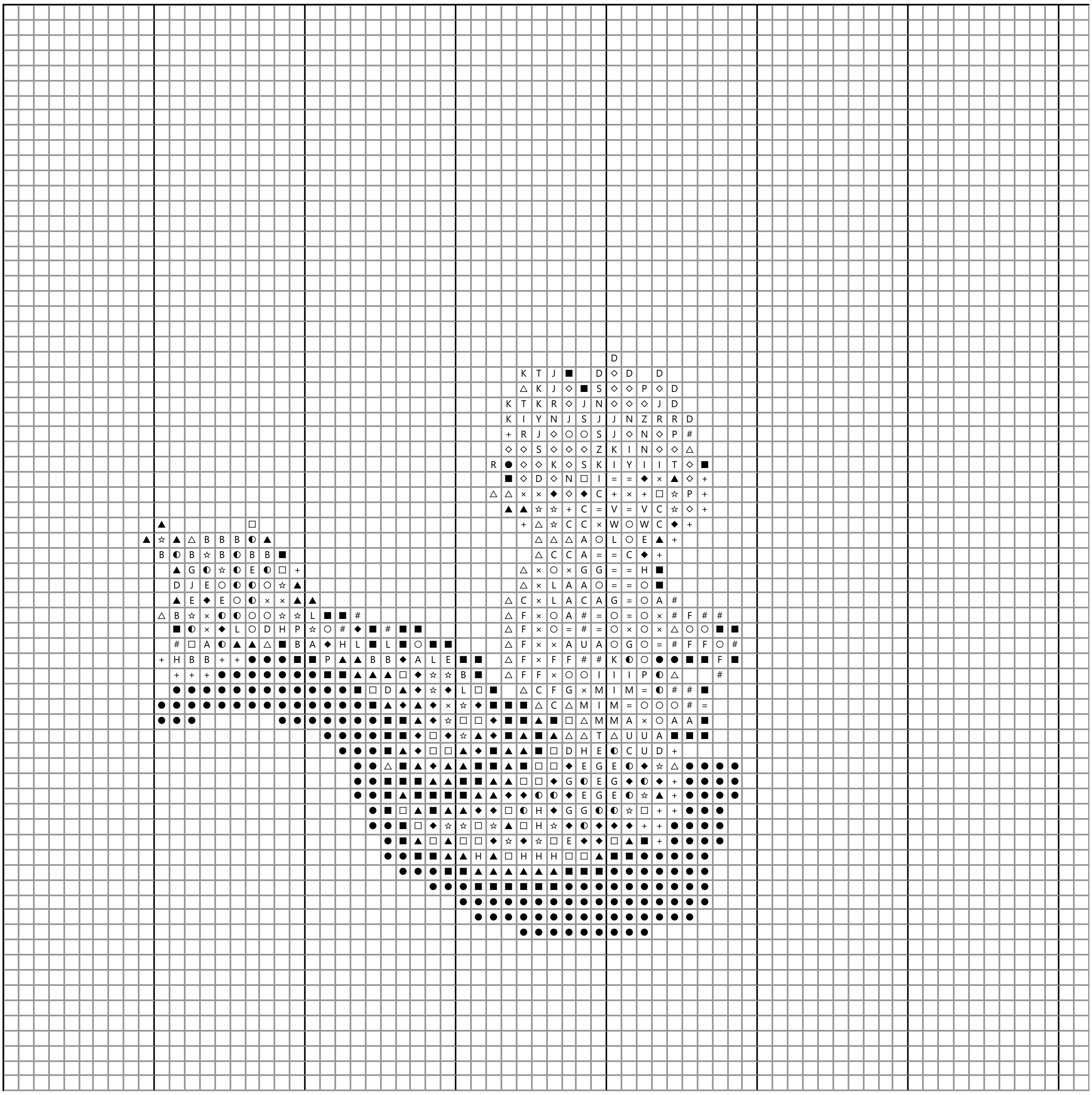
72×72 stitches · k = 20 floss colours · 2.36 mm cells. Read it as you would stitch it: can you tell every symbol apart, and follow a row without losing your place?



Page 5b · The hard case

Merfolk citizen, k = 37 — the richest sprite in the checkout (94 distinct floss, reduced to the cap)

72×72 stitches · k = 37 floss colours · 2.36 mm cells. Read it as you would stitch it: can you tell every symbol apart, and follow a row without losing your place?



Page 6 · Answer key — do not read before page 4

1. H 2. R 3. ☆ 4. G 5. + 6. W 7. ☆ 8. ☆ 9. ◆ 10. C 11. I 12. ○ 13. B 14. U 15. A 16. C 17. ☆ 18. G 19. S 20. ▲ 21. K 22. C 23. K 24. D 25. B 26. P 27. U 28. E 29. V 30. E 31. I 32. ◐ 33. # 34. ☆ 35. C 36. M

The full set, named

● filled circle ■ filled square ▲ filled triangle ◆ filled diamond ○ open circle □ open square △ open triangle ◇ open diamond ☆ open star ◐ half-filled circle **+** plus **×** cross **#** hash **=** equals **A** letter A **B** letter B **C** letter C **D** letter D **E** letter E **F** letter F **G** letter G **H** letter H **I** letter I **J** letter J **K** letter K **L** letter L **M** letter M **N** letter N **P** letter P **R** letter R **S** letter S **T** letter T **U** letter U **V** letter V **W** letter W **Y** letter Y **Z** letter Z

Resolved font: DejaVu Sans, Segoe UI Symbol, Arial available; first available wins.